A SUBSET OF A SPECTRAL SPACE CLOSED IN THE CONSTRUCTIBLE TOPOLOGY IS SPECTRAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that a constructible subset meeting the closure of an irreducible subset contains an open of it. Invoked inside the counted proof (source lines 4658-4660). Shows the generic point of the closure must lie in the subset.
- The statements that the constructible topology is Hausdorff and quasi-compact and that a closed subset of a quasi-compact space is quasi-compact. Invoked inside the counted proof (source lines 4666-4669). Gives quasi-compactness of the subset in the constructible topology.

Source reference. Original lemma lemma-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1879–1879. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-constructible-hausdorff-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 4522–4522. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-hausdorff-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-generic-point-in-constructible: sources/stacks-topology/pinned-original-source.tex, source lines 3048–3048. Navigation pointer only: the counted text refers to the Stacks item lemma-generic-point-in-constructible, which is an existing labeled result of this same chapter and is not part of this card.

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Lemma 0.1. *Let X be a spectral space. Let $E \subset X$ be closed in the constructible topology (for example constructible or closed). Then E with the induced topology is a spectral space.*

Proof. Let $Z \subset E$ be a closed irreducible subset. Let η be the generic point of the closure $\overline{Z}$ of Z in X. To prove that E is sober, we show that $\eta \in E$. If not, then since E is closed in the constructible topology, there exists a constructible subset $F \subset X$ such that $\eta \in F$ and $F \cap E = \emptyset$. By Lemma lemma-generic-point-in-constructible this implies $F \cap \overline{Z}$ contains a nonempty open subset of $\overline{Z}$. But this is impossible as $\overline{Z}$ is the closure of Z and $Z \cap F = \emptyset$.

Since E is closed in the constructible topology, it is quasi-compact in the constructible topology (Lemmas lemma-closed-in-quasi-compact and lemma-constructible-hausdorff-quasi-compact). Hence a fortiori it is quasi-compact in the topology coming from X. If $U \subset X$ is a quasi-compact open, then $E \cap U$ is closed in the constructible topology, hence quasi-compact (as seen above). It follows that the quasi-compact open subsets of E are the intersections $E \cap U$ with U quasi-compact open in X. These form a basis for the topology. Finally, given two $U, U' \subset X$ quasi-compact opens, the intersection $(E \cap U) \cap (E \cap U') = E \cap (U \cap U')$ and $U \cap U'$ is quasi-compact as X is spectral. This finishes the proof. $\square$